

Yao Lu | Curriculum Vitae

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ABOUT ME

I am an undergraduate student from the School of Life Sciences at Peking University, and minor in Artificial Intelligence at Yuanpei College, captivated by nature's most intricate puzzle: the brain. I am fascinated by the biological origins of consciousness and the fundamental principles governing neural computation. Recognizing that these profound questions require interdisciplinary solutions, I actively sought a rigorous background in both biology and AI. My goal is to work at the intersection of these fields—utilizing computational neuroscience to map how the brain processes information, and synergizing these mechanisms with AI to engineer brain-inspired intelligent systems.

RESEARCH INTEREST

Computational Neuroscience, Artificial Intelligence, NeuroAI

EDUCATION

Peking University

B.E. in Life Sciences, B.S. in Artificial Intelligence (Minor)

Beijing, CHN

September 2022- June 2026 (expected)

- Overall GPA: 3.85/4.0
- Minor GPA: 3.84/4.0
- Weighted Average: 91.9/100.0

AWARDS & HONORS

- Leo KoGuan Scholarship, Peking University, 2024
- Second Prize in The Chinese Mathematics Competitions (CMC), Peking University, 2024
- National Scholarship, Peking University, 2023

RESEARCH EXPERIENCE

Closed-Loop Active Learning for Sparse Neural Encoding in Macaque AIT Cortex

2025 - 2026

- The project is under the guidance of Professor Shiming Tang (Shiming Tang).
- Constructed a virtual neural response testbed based on the Maximum Entropy Principle to simulate the highly sparse and specific encoding characteristics of the Macaque AIT cortex.
- Developed a GPU-accelerated closed-loop active learning system utilizing novel algorithms to overcome the "sampling miss" bottleneck for extremely sparse neurons by guiding exploration away from zero-response plains.
- Formulated the closed-loop sampling process as a Markov Decision Process (MDP) and applied Reinforcement Learning (DQN and PPO) to dynamically optimize the retrieval of rare, high-response features.

Brain-Inspired Representation Learning via ResNet-JEPA

2025 - 2026

- The project is under the guidance of Professor Shiming Tang (Shiming Tang).
- Redesigned the standard CNN-JEPA framework to construct a novel ResNet-JEPA model, specifically inspired by the predictive coding theory of the brain.
- Framed self-supervised visual pre-training as a biological predictive process, utilizing the architecture to mimic how the primate visual cortex continuously anticipates incoming sensory inputs in latent spaces.
- Investigated the alignment between the model's latent-space predictive dynamics and the hierarchical representational mechanisms in the ventral visual stream, offering a computational perspective on advanced visual cognition.

RNA editing and RNA biology2024 - 2025

- The project is under the guidance of Professor Chengqi Yi (Yilab).
- Participated in improving the RESTART RNA editing system and exploring suppressor tRNA biology.
- Engineering tRNA with natural and artificial modifications.

Improvement of Equivariant Graph Neural Networks for 3D Atomistic Systems2024 - 2025

- It is a course project under the guidance of Assistant Professor Muhan Zhang (Muhan Zhang).
- Improved the EGNN model based on SOTA model GotenNet, developing the pro-EGNN model.
- The pro-EGNN model combines scalarization-based models and high-degree steerable models, maintaining the ability to encode high-degree geometric features while achieving the computational efficiency of scalarization-based models. Pro-EGNN outperforms the original EGNN on the QM9 dataset for 3D atomistic systems.

Function and Regulation of the TCP Gene Family in Arabidopsis and Soybean2022 - 2023

- The project is under the guidance of Professor Genji Qin (Genji Qin).
- Investigated the regulatory roles of TCP gene family by constructing multiple mutants in Arabidopsis and soybean.
- Focused on the roles of TCP genes in leaf development process.

Undergraduate Honors Program in Biology (UHPB)2023 - 2026

- We hold weekly journal club sessions to share the latest developments in cutting-edge biology and interdisciplinary fields. The topics I have presented include "Integration of Light and Temperature Signals through Phytochrome B (phyB)", "RNA Editing", "Cell Migration under Confinement", and "Computation in Non-Neuronal Cells"

SKILLS

Computation Languages	Python (PyTorch, Scipy, Matplotlib and Scikit-learn), Matlab, R Chinese (native), English (fluent)
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SELECTED COURSES (Full transcript attached)

Mathematics & Physics	Probability Theory and Statistics (98), General Physics (1: 94, 2: A), Adv. Mathematics (1: 93, 2: 91), Intro to Modern Physics II (92), Linear Algebra (87), Intro to Modern Physics I (89)
Computer Science & AI	Cognitive Reasoning (95*), NLP with Deep Learning (93), Machine Learning (92.5*), Intro to Robotics (92), Intro to Computation (91), Math Foundation for AI (90), Early/Mid-Level CV (90), Data Structure and Algorithm (87), Directed Research in AI System (86)
Life Sciences &Interdisciplinary	AI Methods and Applications in Life Sciences (A), Math Modeling in Life Sciences (A), Fundamentals of Network Science and Systems Biology (A)

* denotes graduate-level courses